Factory Predictive Maintenance and Equipment Sustainability Management System

REVIEW 2 – PROJECT PRESENTATION

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Abstract

The **Factory Predictive Maintenance and Equipment Sustainability Management System** is a smart industrial management platform designed to monitor equipment and predict possible failures before they occur. The system collects machine parameters and applies AI/ML techniques to analyze equipment data, identify abnormal conditions, and trigger timely interventions.

Data Collection

Temperature, pressure, vibration, operating hours, and energy consumption

AI/ML Analysis

Identifies abnormal conditions and predicts potential equipment failures

Maintenance & Alerts

Real-time alerts and intelligent maintenance scheduling

Sustainability

Monitors environmental performance and supports sustainable manufacturing

Main goal: Reduce machine downtime, improve equipment reliability, reduce maintenance costs, and support sustainable manufacturing.

Introduction

The Industrial Challenge

Industrial machines are essential for continuous production. Unexpected equipment failures can cause production delays and significant financial losses.

Traditional maintenance methods depend mainly on fixed schedules or manual inspection, and may not identify failures at an early stage.

The Proposed Approach

1. **IoT Sensors**

   Continuously collect machine operating data in real time

2. **AI& Analyse**

   Analyse collected data to predict possible failures before they occur

3. **Integrated Platform**

   Combines Predictive Maintenance + AI/ML + IoT + Sustainability Monitoring

Literature Survey

Research Area	Technologies	Key Findings
Predictive Maintenance in Industry 4.0	IoT, AI, ML	Helps predict equipment failures before they occur
Condition Monitoring	Sensors, Data Analytics	Detects abnormal machine conditions in real time
Smart Manufacturing	IoT, Cloud, AI	Enables real-time monitoring across factory operations
Sustainable Manufacturing	IoT, AI, Analytics	Helps reduce energy and resource wastage
Machine Learning for Maintenance	ML Algorithms	Supports failure prediction and decision-making

Research Gap

Existing solutions often focus on only one area such as monitoring or maintenance. Our proposed system integrates all four dimensions into a single unified platform:

1

Real-Time Monitoring

2

Failure Prediction

3

Maintenance

4

Sustainability

Existing Problems & Objectives

Existing Problems in Industrial Maintenance

- Unexpected machine breakdowns and high equipment downtime
- Manual machine monitoring with difficulty detecting early faults
- High maintenance costs driven by reactive maintenance after failure
- Fixed maintenance schedules that may be inefficient
- Excessive energy consumption and resource wastage
- Lack of centralized equipment information
- Delayed alerts and slow decision-making

System Objectives

1 Predict Equipment Failures

Use sensor data and AI/ML techniques to identify abnormal conditions and predict potential equipment failures

2 Improve Maintenance Efficiency

Provide real-time alerts and maintenance scheduling to reduce downtime and improve equipment reliability

3 Improve Sustainability

Monitor energy consumption and environmental performance to reduce resource wastage and support sustainable factory operations

Proposed Solution

Our proposed system provides a **centralized smart factory platform** for equipment management. The core idea is to predict problems **before equipment failure** and use the collected data to improve both maintenance and sustainability.

Complete solution flow: industrial machines → IoT sensors collecting data → cloud processing → AI analysis → alerts and notifications → maintenance scheduling → healthy equipment.

Key Differentiators

Proactive Approach

Predicts failures before they occur rather than reacting after breakdown

Unified Platform

Single system covering monitoring, prediction, maintenance, and sustainability

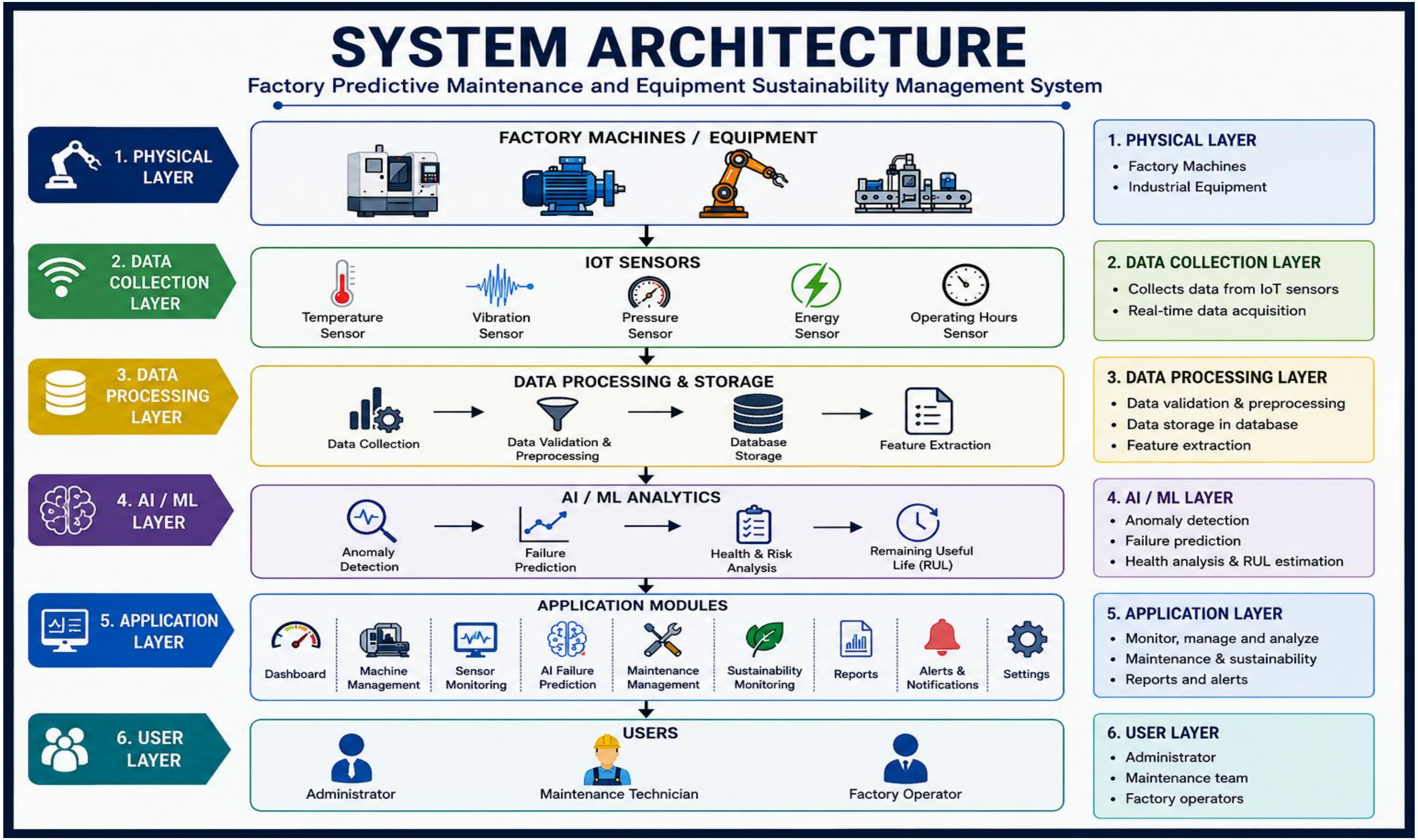
AI-Driven Intelligence

Machine learning continuously improves prediction accuracy over time

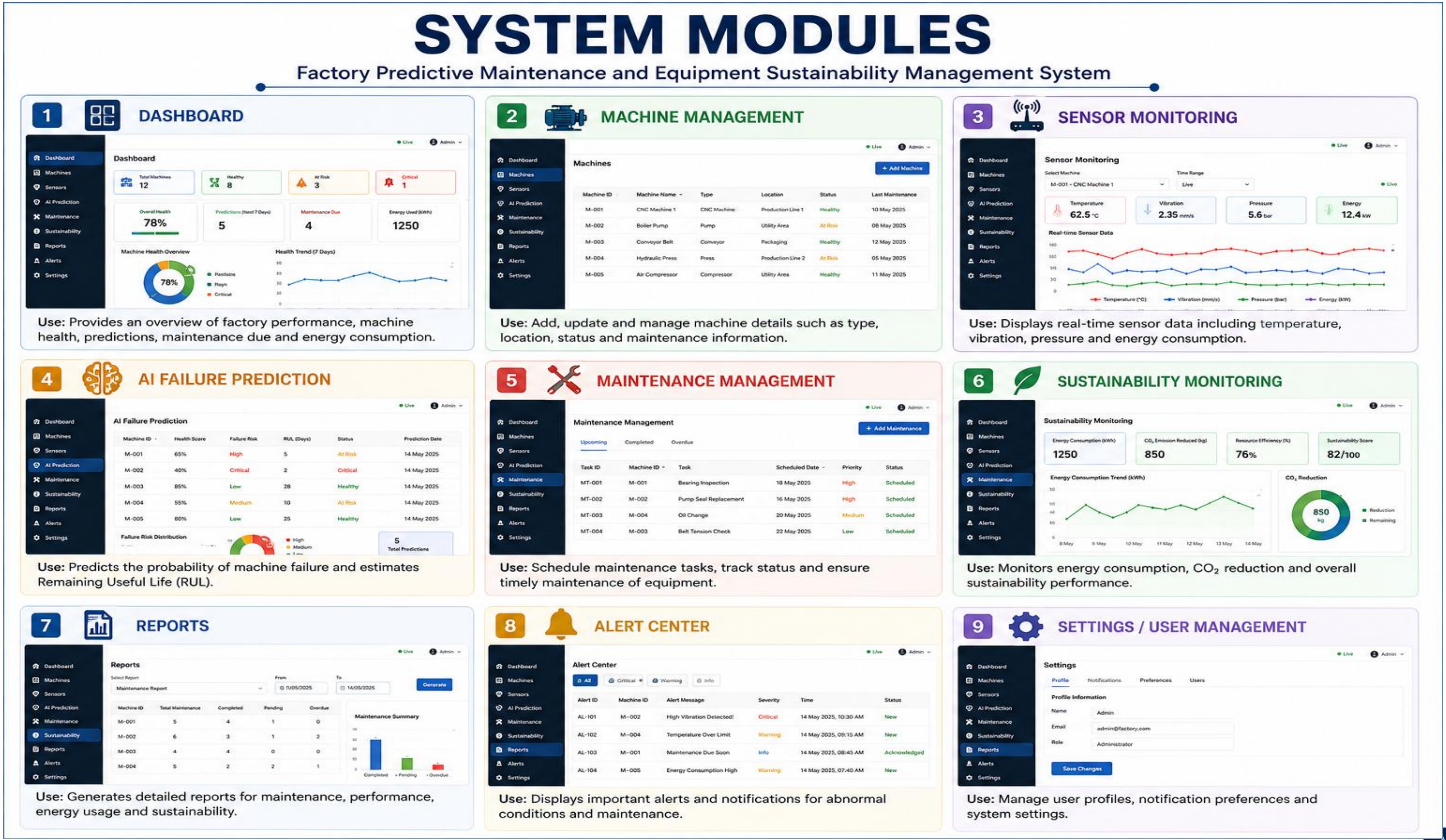
Actionable Insights

Dashboard and reports enable data-driven decision-making

System Architecture

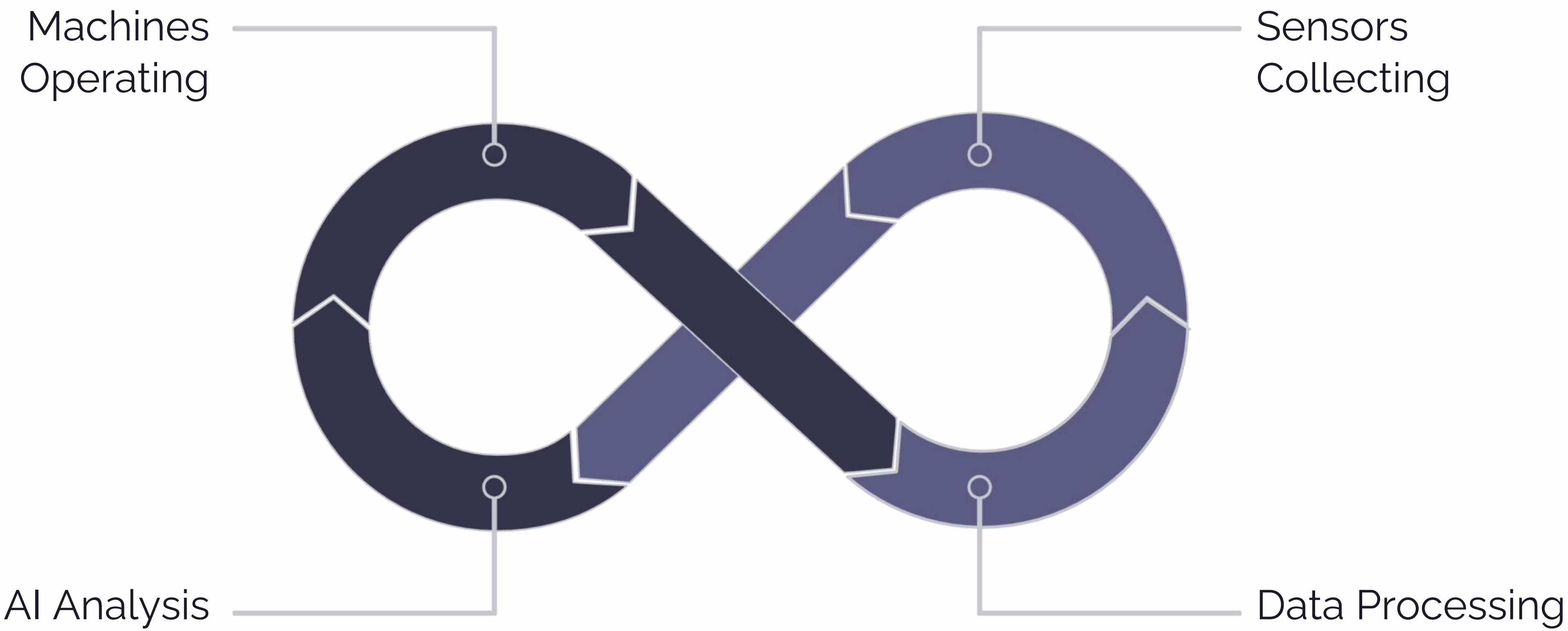


System Modules



System Workflow

The end-to-end workflow describes how data flows from physical machines through AI analysis to actionable maintenance and sustainability outcomes.



Summary & Conclusion

The **Factory Predictive Maintenance and Equipment Sustainability Management System** addresses the critical gap in existing industrial maintenance approaches by delivering a unified, AI-powered platform that spans the full equipment management lifecycle.

What the System Delivers

- Early failure prediction using IoT sensor data and ML algorithms
- Real-time alerts and intelligent maintenance scheduling
- Centralized dashboard covering all nine functional modules
- Sustainability monitoring for energy and CO_2 performance
- Automated reports for data-driven decision-making

Advantages

Reduces unexpected equipment breakdowns

Reduces machine downtime

Provides early fault detection

Provides real-time machine monitoring

Improves equipment reliability

Provides centralized factory information

Helps monitor energy consumption

Supports sustainable manufacturing

Improve maintenance planning

Conclusion

IoT

AI/ML

Real-Time Monitoring

Predictive Maintenance

Sustainability

Short explanation:

"Our system uses IoT and AI to monitor machines, detect problems early, and predict maintenance. This reduces downtime, improves machine reliability, and supports sustainable smart manufacturing."

Key outcomes

- Early identification of potential equipment problems
- Efficient maintenance scheduling
- Reduced downtime
- Improved equipment reliability

From traditional maintenance

Reactive, manual, and harder to optimize.

To smart manufacturing

Connected, predictive, efficient, and sustainable.

References

Academic Research

1. Mallioris, P., Aivazidou, E., & Bechtsis, D. — *Predictive Maintenance in Industry 4.0: A Systematic Multi-Sector Mapping*.
2. *Paradigm Shift for Predictive Maintenance and Condition Monitoring from Industry 4.0 to Industry 5.0*.

Research Focus Areas

1. Research studies on IoT-enabled Predictive Maintenance and Sustainable Manufacturing.
2. Research studies on Artificial Intelligence and Machine Learning for Industrial Equipment Monitoring.
3. Industry 4.0 research on Smart Manufacturing, IoT and Data Analytics.

Project Resources

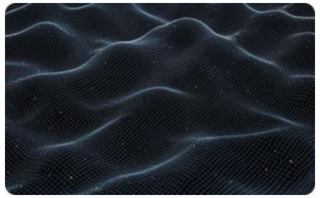
1. Project Website: Factory Predictive Maintenance and Equipment Sustainability Management System.

Academic Sources

Scholarly references supporting predictive maintenance and Industry 4.0 research.

Technical Knowledge

Documentation and research themes covering IoT, AI/ML, and smart manufacturing.



Project Resource

Reference material and website resources for the system implementation.

THANK YOU

THANK YOU FOR YOUR VALUABLE TIME

QUESTIONS & ANSWERS?

Presenters

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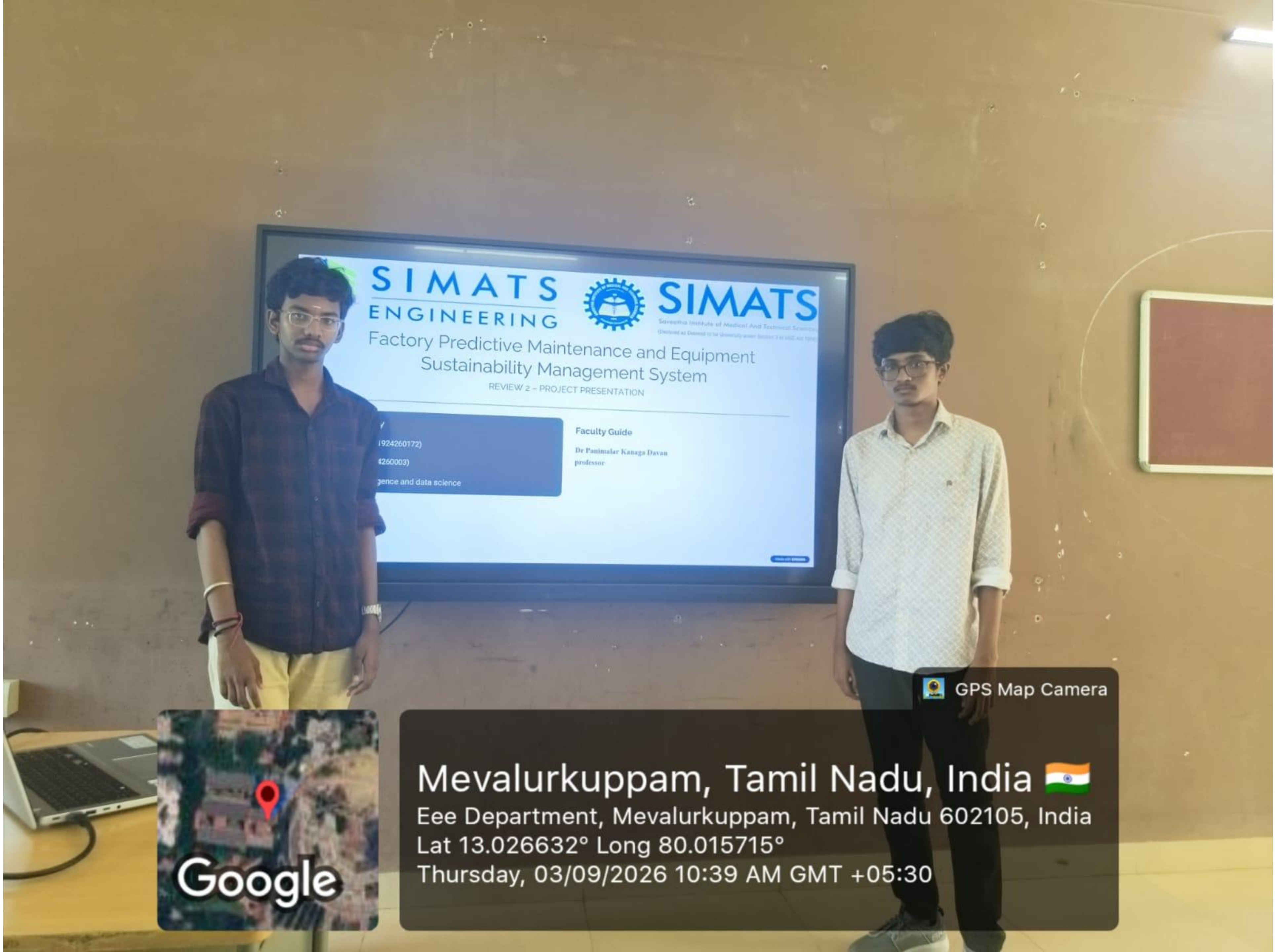
FACTORY PREDICTIVE MAINTENANCE AND EQUIPMENT SUSTAINABILITY MANAGEMENT SYSTEM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